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Hades: A Hardware-Resident Substrate for Energy Admission in Intermittent Computing

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The hidden cost of tiny sensors

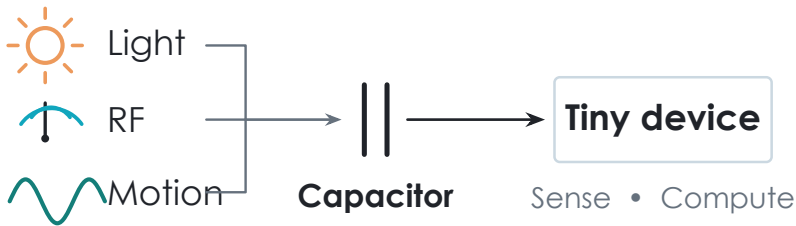
IoT scales. So does battery maintenance.



Battery replacements add maintenance, e-waste, and carbon footprint at IoT scale.

What if sensors harvested energy?

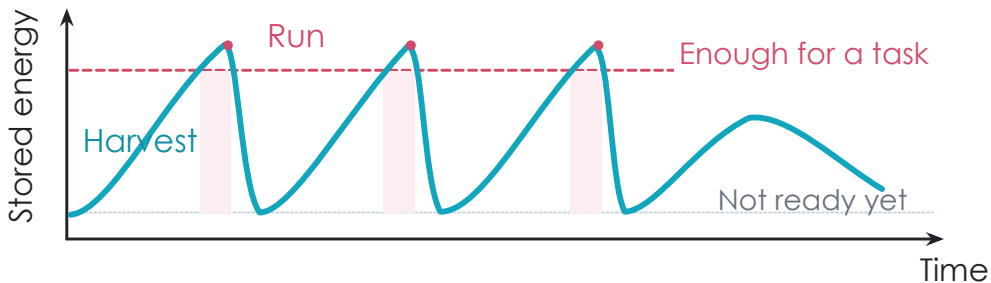
Light, radio waves, and motion can power tiny devices.



The catch: energy arrives in unpredictable bursts.

Energy arrives. Tasks wait.

A capacitor charges. A task runs. Then it waits.



Who decides when the next task can safely start?

Related work: different goals

Prior work improves different parts of batteryless operation.

Software-mediated admission
Mementos • Dewdrop
Runtime and scheduling

Hardware-assisted admission
Failure Sentinels • CapDYN
Monitoring and buffering

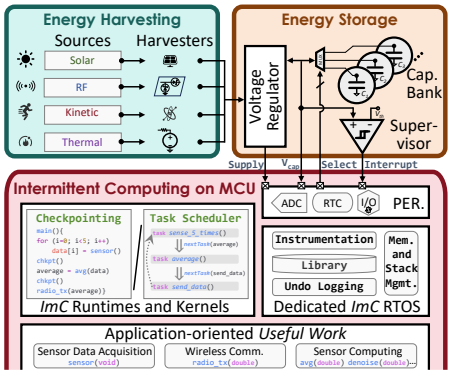
Post-admission execution
Clank • ReplayCache
Persistence and recovery

Pre-admission energy intake
AmbiMax • InfiniteEn • MPPT
Harvesting and buffering

Let us zoom in on task admission.

How does a task get admitted?

A voltage supervisor wakes the microcontroller unit (MCU).



1. Harvest and store
Capacitor banks collect energy.

2. Hardware wakeup
Supervisor triggers the MCU.

3. Software admission

Sample storage via ADC

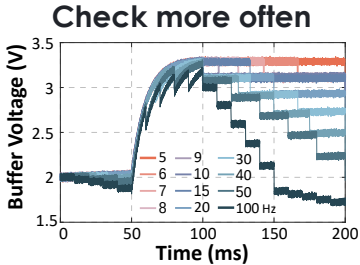
Evaluate available energy

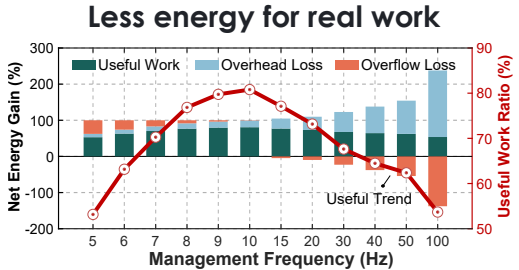
Choose the next task

Execute or return to sleep

Checking energy costs energy

Problem: Frequent MCU-side checks improve responsiveness, but each wakeup costs energy.

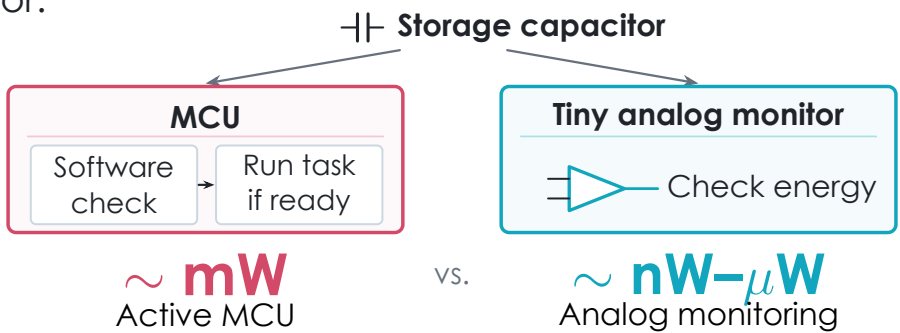




More checking can mean less energy for useful tasks.

The processor is power-hungry

An active MCU uses far more power than a tiny analog monitor.

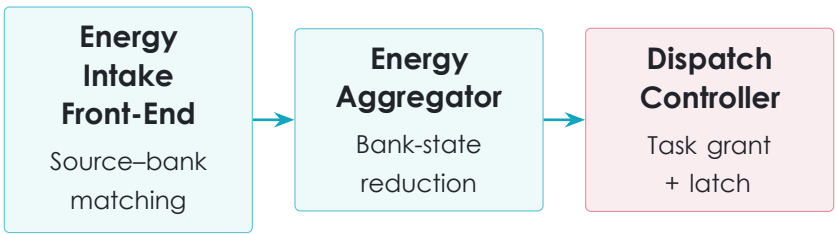


Why wake an mW-class MCU just to check energy?

Meet Hades: harvest more, spend less

Insight: Keep energy admission in tiny always-on hardware; wake the MCU for useful tasks.

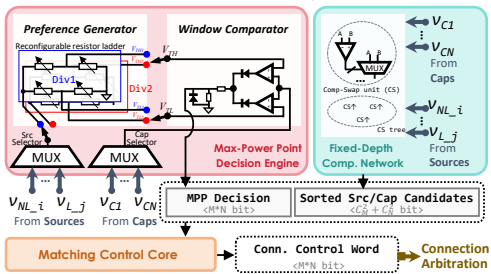
Always-on mixed-signal substrate



Two benefits: Better energy intake. Less management energy.

Near-MPPT, without costly tracking

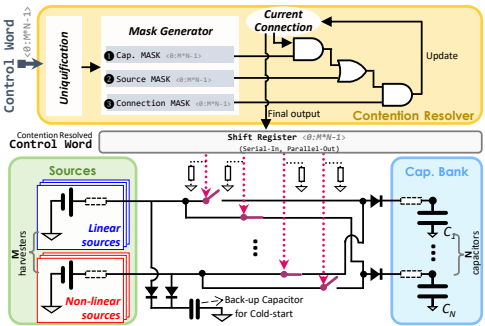
Problem: Digital MPPT repeatedly samples and retunes the load, adding control overhead at μW inputs.



Insight: Match source voltage windows to existing bank voltages. Near-MPPT capture, without continuous digital tracking.

Good matches can conflict

Problem: Many feasible source–bank pairs compete for the same connections.



1. Uniquify requests
Order candidates.

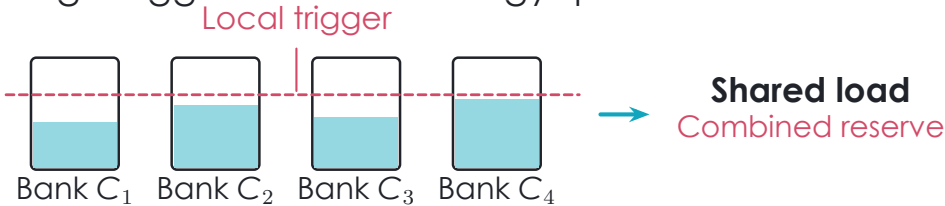
2. Generate three masks
Cap: no shared bank
Source: one bank per source
Connection: respect active links

3. Resolve and commit
Resolve against active connections.
Shift, then commit atomically.

Insight: Masks select valid links; one stable switch update.

One bank looks low. Together, enough.

Local voltage triggers can miss energy spread across banks.



$$E_{\text{stored}} = \frac{1}{2} \sum_{i=1}^{N} C_i V_{C,i}^2$$

Question: can hardware estimate this sum without ADC scans?

Let the circuit compute energy

ADCs cost power. Comparator ladders lose detail.

Energy stored in one bank

$$E_i = \frac{1}{2} C_i V_{C,i}^2$$

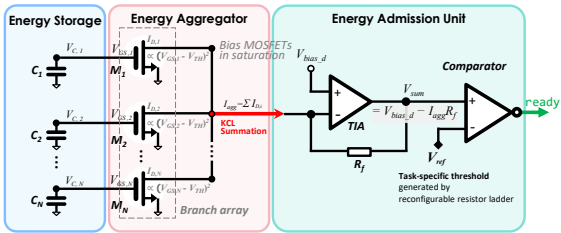
same V^2 form

Current from one MOS branch

$$I_{D,i} \approx K_i \left[(V_{GS,i} - V_{TH})_+ \right]^2$$

Voltage: $V_{GS,i}$ follows $V_{C,i}$ (with bias)

Weight: $K_i \propto C_i$

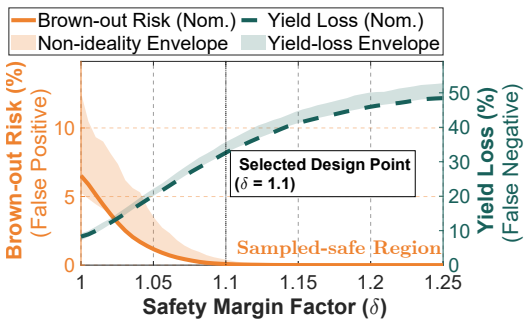


1. **MOS square law**
   Voltage → current
2. **Transistor sizing**
   $K_i \propto C_i$
3. **KCL summation**
   $I_{agg} = \sum_i I_{D,i}$

$$I_{agg} = \sum_i I_{D,i} \longrightarrow V_{sum} \xrightarrow{V_{ref,t}} \texttt{ready}$$

A fast comparison needs a safe threshold

Problem: An analog energy proxy can over-admit a task or miss a safe execution window.



Insight: Calibrate a task-specific threshold offline.

At the edge

Profile energy and droop.

On the device

Load $V_{\text{ref},t}$.

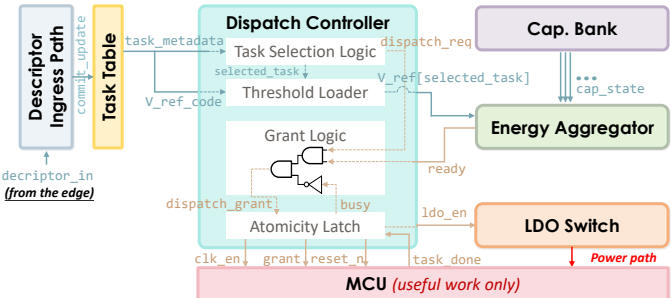
Margin $\delta = 1.1$

0.38%

Stressed FP, 95th percentile

Different tasks. Different energy budgets.

Problem: The same stored energy cannot support every task.
Insight: Profile offline; compare against a task-specific threshold.



1. Profile at the edge
Energy + droop budget.

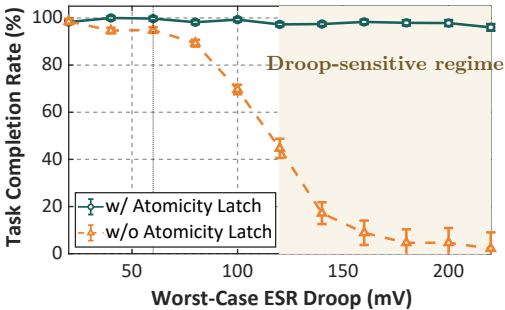
2. Store the threshold
Bind task ID to $V_{\text{ref},t}$.

3. Dispatch in hardware
Compare, grant, then wake MCU.

$$E_{\text{iso},t} \xrightarrow{\text{calibration}} V_{\text{ref},t} \qquad \texttt{ready} \Leftrightarrow V_{\text{sum}} \leq V_{\text{ref},t}$$

Once it starts, keep the window open

Problem: MCU current draw makes the capacitor voltage fall.
Insight: Latch the admission decision until the task completes.



1. Qualify the start
req ∧ ready ∧ ¬busy

2. Latch the grant
Ignore expected droop.

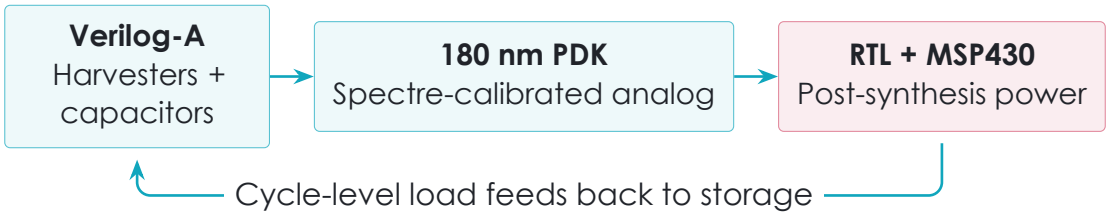
3. Close at completion
`task_done` clears busy.

Hold the grant despite voltage droop; release at completion.

The latch holds admission, not persistent state or recovery.

How we evaluate Hades

Method: Closed-loop modeling couples harvested energy, analog circuits, and MCU activity.



12 embedded workloads
AES, FFT, SHA, etc.

4 energy sources
Solar, RF, thermal, motion

180 nm PDK
Calibrated circuit models

Always on, with little overhead

Question: Is a permanently active admission circuit cheap enough?

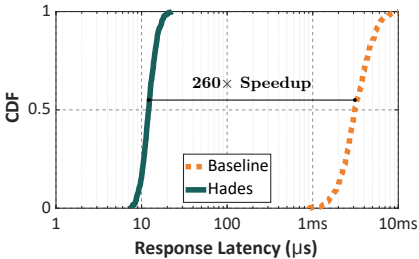
0.29%	**542 nW**	**13.13 pJ**	**1.05 μs**
evaluated SoC area	always-on power	dynamic energy / decision	decision response

Software: **2.51 μJ / attempt** $\longrightarrow$ >98% lower modeled admission energy

Act fast on weak energy

Problem: MCU-side checks may miss short energy bursts.

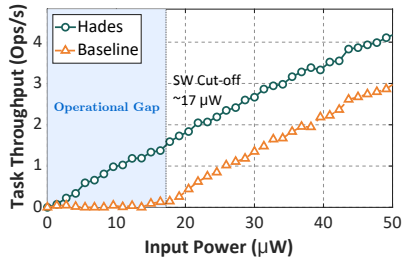
Burst-to-execution latency



12 μs vs. 3.2 ms

Median $\approx 260\times$ faster

Minimum viable input power



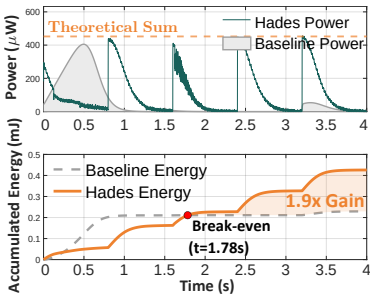
$\sim 2\ \mu W$ vs. $\sim 17\ \mu W$

Evaluated workload/trace settings

Result: Faster admission makes **brief, weak energy bursts** usable.

Better matching. More energy stored.

Question: Does analog matching actually store more energy?



1.9×

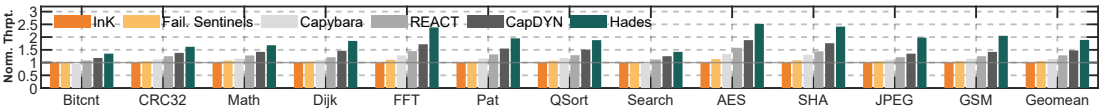
Energy delivered to storage

Solar + RF, 4-second trace

Result: More harvested energy reaches storage, without continuous MPPT tracking.

More work gets done

System impact: Across 12 workloads, improved admission translates into more completed tasks.



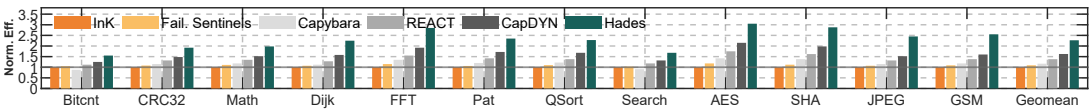
2.2×
geometric mean vs. InK

1.35×
geometric mean vs. CapDYN

2.5×
up to vs. InK

More tasks from every joule

System impact: Better intake and cheaper admission turn more energy into useful work.



2.4×

geometric-mean tasks/J over InK

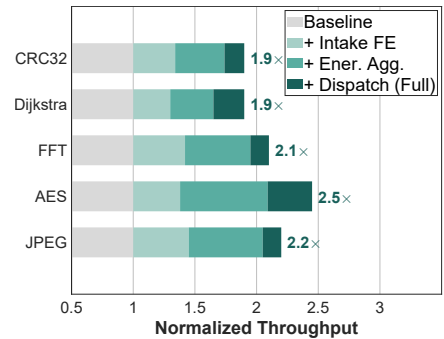
Why it works

Collect more · Check less

Dispatch only when ready

Why all three blocks matter

Evidence: Removing a block loses part of the gain; the three mechanisms complement each other.



Front-End
$1.38\times$

+ Aggregator
$1.9\times$

+ Dispatch
up to $2.45\times$

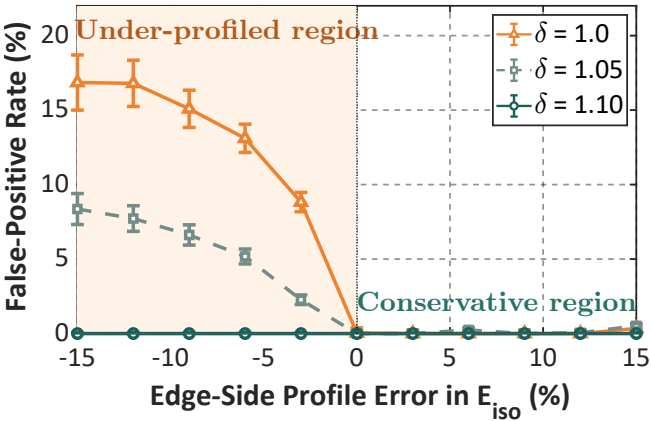
Takeaway

Hades moves admission off the MCU, from repeated software checks to always-on hardware.

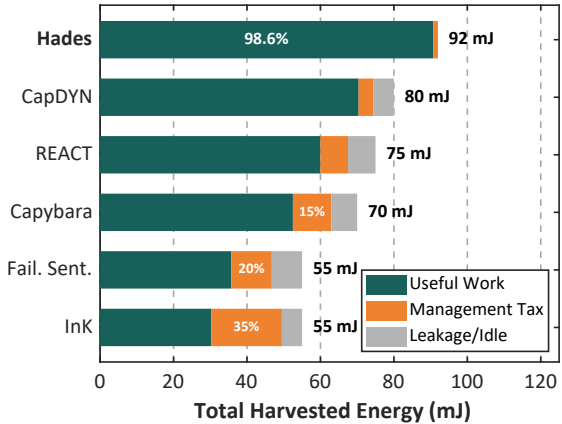
- **Energy Intake:** near-MPPT capture with low control cost
- **Energy Aggregator:** compute a distributed reserve in analog
- **Dispatch Controller:** grant and hold a safe task window

Let hardware decide. Let the MCU work.

Backup: threshold sensitivity to profile uncertainty



Backup: the modeled energy budget breakdown



Backup: storage topology sensitivity

